

Assistance Needs Observed in Human–Human Coassembly

851 annotated assistance events



DETAILED EVENT CATEGORIES

- **PHYSICAL SUPPORT**
Part Out of Reach — 172/851 (20.2%)
Workpiece Stabilization — 36/851 (4.2%)
- **PART SEARCH & TASK DECISIONS**
Unable to Locate Part — 124/851 (14.6%)
Uncertain About Next Step — 123/851 (14.5%)
- **ASSEMBLY ERROR CORRECTION**
Incorrect Part Selection — 96/851 (11.3%)
Assembly-Order Violation — 63/851 (7.4%)
Incorrect Part Orientation — 63/851 (7.4%)
Incorrect Screwdriver / Bit — 57/851 (6.7%)
Missed Fastener — 22/851 (2.6%)
- **CONVERSATIONAL SUPPORT**
Confirmation / Affirmation — 52/851 (6.1%)
Other Conversational Support — 43/851 (5.1%)